Homework #2

CS5402 — Intro To Data Mining

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1 Association Rules

Problem Statement. *Compute the coverage (i.e., support) of each item set listed below.*

drinkPref = pepsi 2

married = no, pet = dog 3

sportPref = football, musicPref = classical, drinkPref = tea 0

Problem Statement. *Write down every association rule that could be generated from the item set listed below, regardless of whether or not there are actually any instances of that rule in our given dataset.*

$$\begin{aligned}
\textbf{married} = \text{no} &\implies \textbf{pet} = \text{dog} \\
\textbf{pet} = \text{dog} &\implies \textbf{married} = \text{no} \\
_ &\implies \textbf{married} = \text{no and } \textbf{pet} = \text{dog}
\end{aligned}$$

Problem Statement. *Compute the accuracy (i.e., confidence) of each rule listed below. Express accuracy as a fraction (e.g., 2/3, 2/2, etc.), NOT as a decimal number (e.g., 0.67, 1.0, etc.).*

With the case *If pet = dog then income = middle*, we get as follows:

$$\text{Accuracy} = \frac{1}{4}$$

With the case *If married = no and sportPref = football, then pet = dog and musicPref = rock*, we get as follows:

$$\text{Accuracy} = \frac{2}{3}$$

With the case *If _, then drinkPref = coke and married = yes*, we get as follows:

$$\text{Accuracy} = 0$$

Problem Statement. *In each rule listed below, specify whether any condition(s) in the antecedent (i.e., the "if' part) can be dropped.*

If married = no and sportPref = football then pet = dog and drinkPref = coke No.

If married = yes and pet = cat then musicPref = country Yes, **married** = yes can be dropped.

2 RICO

Problem Statement. *Using the Rule Induction from Coverings algorithm discussed in class, find each of the partitions specified below.*

$$\begin{aligned}
\{a\}^* &= \{\{x_1, x_5\}, \{x_2, x_3, x_4\}\} \\
\{c\}^* &= \{\{x_1\}, \{x_2, x_3\}, \{x_4\}, \{x_5\}\} \\
\{d\}^* &= \{\{x_1, x_2\}, \{x_3, x_4\}, \{x_5\}\} \\
\{d,\ e\}^* &= \{\{x_1\}, \{x_2\}, \{x_3, x_4\}, \{x_5\}\}
\end{aligned}$$

Problem Statement. *Suppose that we are looking for a covering for decision attribute f which has partition {f}* = {{x1, x2}, {x3, x4, x5}}.*

- No, the block $\{x_1, x_5\}$ is not a subset of any blocks in $\{f\}^*$.
- No, the block $\{x_2, x_3\}$ is not a subset of any blocks in $\{f\}^*$.
- Yes, all blocks are subsets of all blocks in $\{f\}^*$.
- No, because $\{d,\ e\}^*$ is not minimal.

Problem Statement. *Find a covering for f. If you found one in part b that worked, just use one of those. Using your covering, give a set of rules for decision attribute f.*

$$\begin{aligned}
d = 0 &\implies f = True \\
d = 1 &\implies f = False \\
d = 2 &\implies f = False
\end{aligned}$$

3 KD-Tree

Problem Statement. *Consider the dataset given below where the decision attribute is the one labeled decision. Build a kd-tree where k = 3. No partial credit will be given unless you show your work!*

Sorting the tuples by x, we get as follows:

$$[(1, 7, 9), (2, 5, 8), (3, 6, 7), (4, 4, 0), (5, 3, 4), (6, 1, 6), (7, 2, 2)]$$

Seeing a our median = 4, we partition as follows:

$$[(1, 7, 9), (2, 5, 8), (3, 6, 7)] \quad [(4, 4, 0), (5, 3, 4), (6, 1, 6), (7, 2, 2)]$$

Sorting these tuples by y, we get as follows:

$$[(2, 5, 8), (3, 6, 7), (1, 7, 9)] \quad [(6, 1, 6), (7, 2, 2), (5, 3, 4), (4, 4, 0)]$$

With our medians as follows, we get as follows:

$$\begin{array}{llll} \text{median} = 6 & & \text{median} = 2.5 & \\ [(2, 5, 8)] & [(3, 6, 7), (1, 7, 9)] & [(6, 1, 6), (7, 2, 2)] & [(5, 3, 4), (4, 4, 0)] \end{array}$$

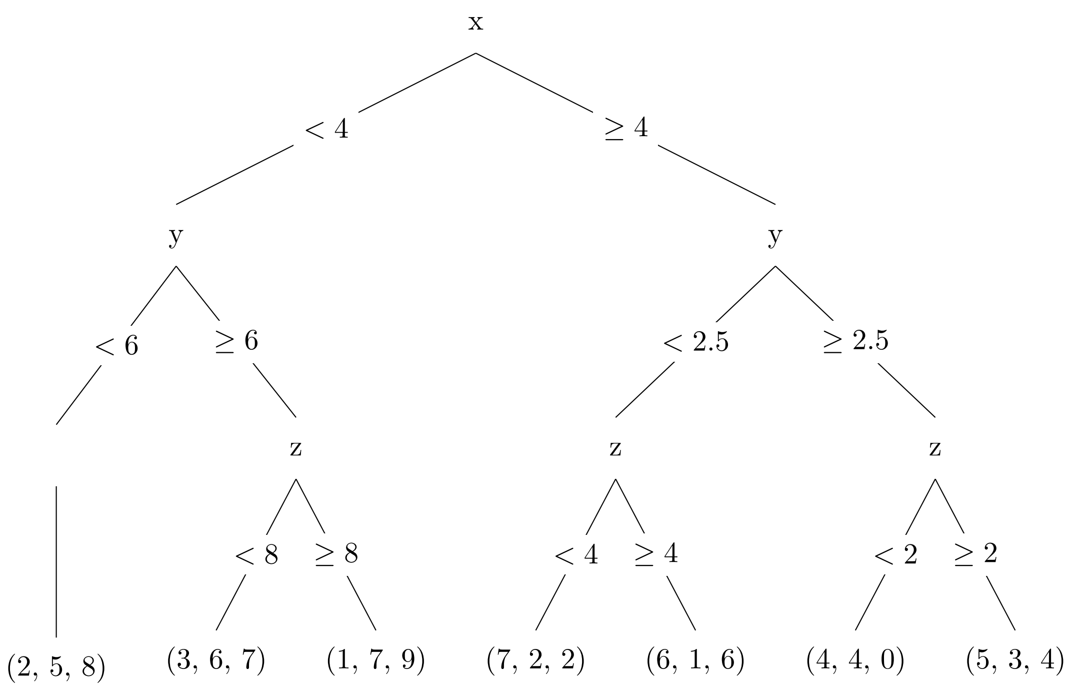
Sorting these tuples by z, we get as follows:

$$[(2, 5, 8)] \quad [(3, 6, 7), (1, 7, 9)] \quad [(7, 2, 2), (6, 1, 6)] \quad [(4, 4, 0), (5, 3, 4)]$$

With our medians as follows, we get as follows:

$$\begin{array}{lllllll} \text{median} = 8 & \text{median} = 8 & & \text{median} = 4 & & \text{median} = 2 & \\ [(2, 5, 8)] & [(3, 6, 7)] & [(1, 7, 9)] & [(7, 2, 2)] & [(6, 1, 6)] & [(4, 4, 0)] & [(5, 3, 4)] \end{array}$$

To produce the following KD-Tree:



4 Winnow

Instance	**Calculation**	**Predicted**	**Actual**	**Resultant Weights**
x_1	$2 \times 1 + 2 \times 0 + 2 \times 1 = 4 > 2$	1	0	$w_a = 1, w_b = 2, w_c = 1$
x_2	$1 \times 0 + 2 \times 1 + 1 \times 0 = 2 \leq 2$	0	1	$w_a = 1, w_b = 4, w_c = 1$
x_3	$2 \times 1 + 4 \times 1 + 2 \times 0 = 6 > 2$	1	1	$w_a = 1, w_b = 4, w_c = 1$
x_4	$1 \times 1 + 4 \times 0 + 1 \times 1 = 2 \leq 2$	0	0	$w_a = 1, w_b = 4, w_c = 1$